

HEAVEN'S LIGHT IS OUR GUIDE



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF MECHATRONICS ENGINEERING, 21Series

Course No: MTE 3102

Course Title: Control Systems Sessional

Experiment No: 01

Name of the Experiment: System Response of a State Space Model using MATLAB & Simulink

Submitted To:

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We know, a state space model is represented by:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{z}$$

Let the state-space matrices be defined as follows:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = [1 \quad 0], \quad \mathbf{D} = [0]$$

MATLAB Code:

```
clc;
clear all;
close all;
% Define State-Space Matrices
A = [0 1; -2 -3];
B = [0; 1];
C = [1 0];
D = 0;
% Create State-Space System
sys = ss(A, B, C, D);

% Step Response Analysis
figure;
step(sys); % Step response
title('Step Response');
grid on;

% Bode Diagram Analysis
figure;
bode(sys); % Bode plot
title('Bode Diagram');
grid on;
```

Result:

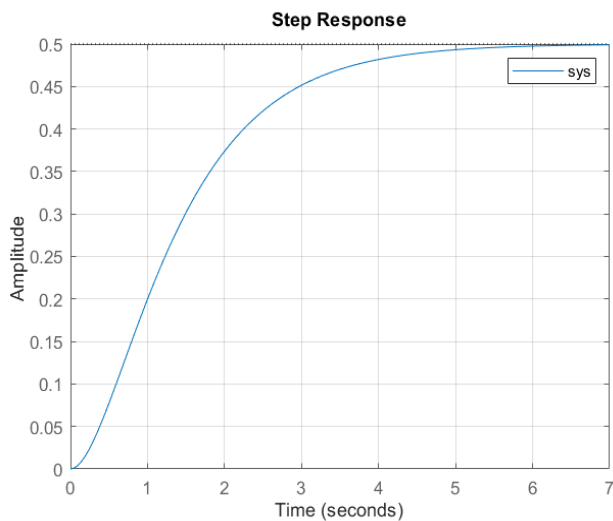


Figure 1: Step Response (MATLAB)

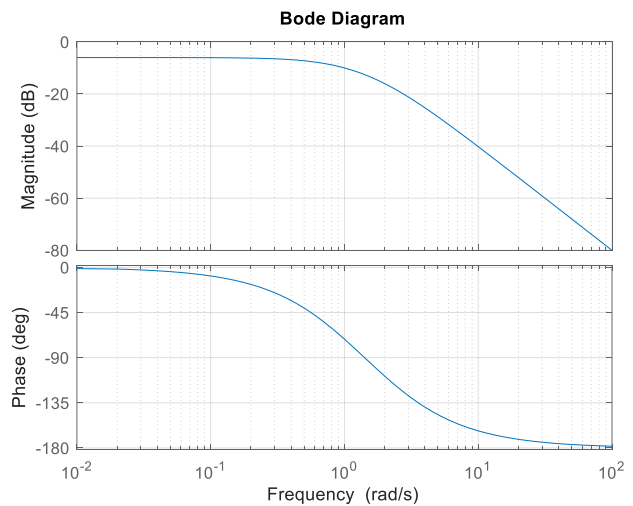


Figure 2: Bode Diagram (MATLAB)

Simulink:

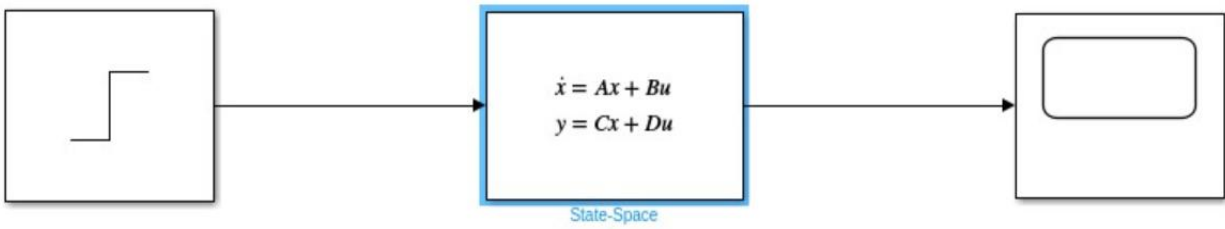


Figure 3: Simulink Model Representation

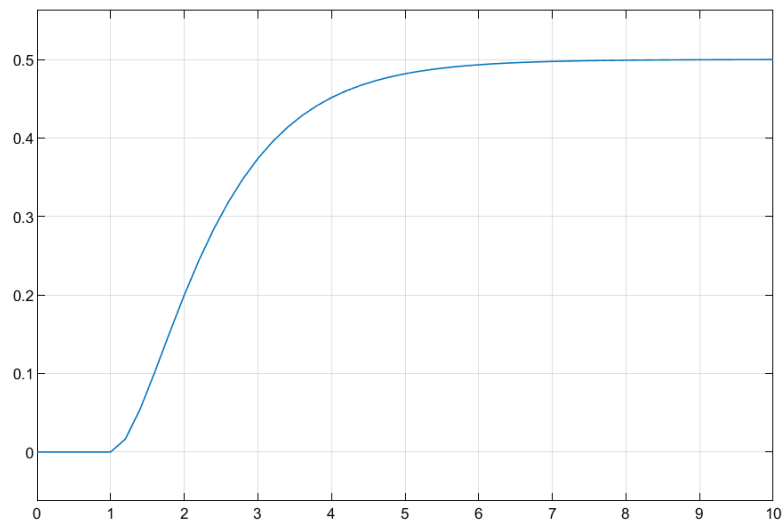


Figure 4: Step Response (Simulink)

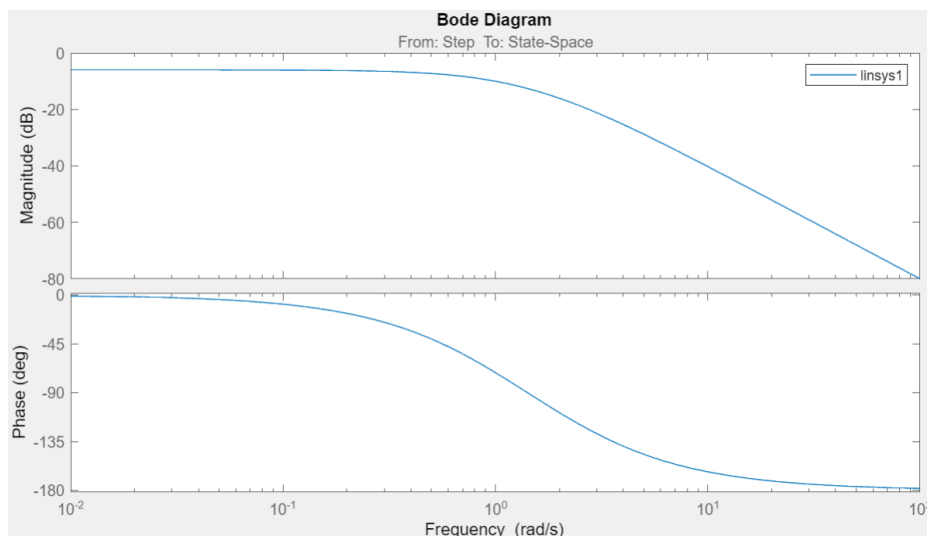


Figure 5: Bode Diagram (Simulink)